Maths - 2

Week - 2

L1 Determinants (Part - 3)

- recall determinants & cofactors.
- we can expand along any row or column.

$$\det(A) = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} M_{ij} \quad \text{for a fixed } i$$

↳ expansion along the i^{th} row.

$$= \sum_{i=1}^{n} (-1)^{i+j} a_{ij} M_{ij} \quad \text{for a fixed } j$$

↳ expansion along the j^{th} column.

- Properties -

1. determinant of a product is the product of the determinants

$$\det(AB) = \det(A)\det(B)$$

$$\det(A^n) = \det(A)^n$$

$$\det(A^{-1}) = \frac{1}{\det(A)} = \det(A)^{-1}$$

$$\det(AB) = \det(BA)$$

$\det(A^T A) = \det(A)^2$

2. Switching 2 rows or columns changes the sign.
3. Adding multiples of a row to another row leaves the determinant unchanged.
4. Scaler multiplicaⁿ of a row by a constant t multiplies the determinant by t.

Warning → $\det(tA_{n\times n}) = t^n \det(A)$

- det. of a matrix has a 0 row/col. is always 0.
- if one row/col. is linear combinaⁿ of other row/col., then det. is 0.
- scaler multiplicaⁿ of a row by a constant t multiplies the det. by t.

L2 Cramer's Rule

- for invertible matrices —

$$\begin{aligned} 4x_1 - 3x_2 &= 11 \\ 6x_1 + 5x_2 &= 7 \end{aligned} \Rightarrow A = \begin{bmatrix} 4 & -3 \\ 6 & 5 \end{bmatrix}$$

$$b = \begin{bmatrix} 11 \\ 7 \end{bmatrix}$$

unique solⁿ does exist —

$x_1 = 2$, $x_2 = -1$

- calculate $\det(A) \Rightarrow 38$

$$A_{x_1} = \begin{array}{cc} b & \\ \downarrow & \\ \end{array} \begin{bmatrix} 11 & -3 \\ 7 & 5 \end{bmatrix} \qquad A_{x_2} = \begin{bmatrix} 4 & 11 \\ 6 & 7 \end{bmatrix}$$

$\det(A_{x_1}) = 76$ $\qquad$ $\det(A_{x_2}) = -38$

$$x_1 = \frac{\det(A_{x_1})}{\det(A)}$$

$$= \frac{76}{38} = 2$$

$$x_2 = \frac{\det(A_{x_2})}{\det(A)} = \frac{-38}{38} = -1$$

- the Cramer's rule is applied only on invertible matrices.

$\hookrightarrow \det(A) \neq 0$.

- coefficient matrix must be a sq. matrix + invertible. Then Cramer's rule can be defined for $n \times n$ matrices.

L3 Sol^{ns} to the Sys. of Linear Eq^{ns} with an invertible coefficient matrix

- recall the square matrix & invertible matrix.

$$AA^{-1} = A^{-1}A = I_{n\times n}$$

$A^{-1} \Rightarrow$ is the inverse matrix of A. (it is a unique matrix)

- inverse of A exists
  $\Rightarrow$ det (A) has to be non-zero.

for a 2x2 matrix, $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$\boxed{A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}}$$

- adjugate of the square matrix –

$$C_{ij} = (-1)^{i+j} M_{ij}$$

$$\boxed{\operatorname{adj}(A) = C^T}$$

$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 8 \\ 5 & 6 & 0 \end{bmatrix}$ Calculate all the minors

$\det(A) = 2$

$C = \begin{bmatrix} -48 & 40 & -10 \\ 18 & -15 & 4 \\ 10 & -8 & 2 \end{bmatrix}$

$\Downarrow C^T = \operatorname{adj}.$

$\Rightarrow \operatorname{adj}.(A) = \begin{bmatrix} -48 & 18 & 1 \\ 40 & -15 & - \\ -10 & 4 & 2 \end{bmatrix}$

$$\boxed{A^{-1} = \frac{1}{\det(A)} \operatorname{adj.}(A)}$$

for A a $n \times n$ matrix, $\det(A) \neq 0$, then only A^{-1} exists.

- Now, consider the sys. of linear eq^{ns} $Ax = b$, where A (the coefficient matrix) is invertible.

$$Ax = b \Rightarrow A^{-1}Ax = A^{-1}b$$

$$\boxed{x = A^{-1}b}$$

Eg. $A = \begin{bmatrix} 8 & 8 & 4 \\ 12 & 5 & 7 \\ 3 & 2 & 5 \end{bmatrix}$ $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ $b = \begin{bmatrix} 1960 \\ 2215 \\ 1135 \end{bmatrix}$

$\det(A) = -188 \neq 0 \Rightarrow$ A is invertible.

minor $\Rightarrow C_{ij} \Rightarrow \operatorname{adj.}(A) \Rightarrow A^{-1} = \frac{1}{\det(A)} \operatorname{adj.}(A)$

$x = A^{-1}b$

- A system of linear eq^{n} is homogenous if all of the constant terms is 0. $b = 0$.

matrix $\Rightarrow Ax = 0$, n eq^{n}, n unknown.

→ unique sol^{n} 0, if $\det \neq 0$

→ ∞ sol^{n}, if $\det = 0$.

L4 The Echelon Form

- recall the system of linear equations
- a solu^n is an assignment of values for x so that equations are satisfied (holds true)

Eg. $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}$ $B = 0$

$Ax = 0$

$x_1 + 2x_3 = 0$
$x_2 + 3x_3 = 0$ $\Rightarrow$ $x_1 = -2x_3$
$x_2 = -3x_3$

if, $x_3 = c \Rightarrow x_1 = -2c, \; x_2 = -3c$

set of solu^n $\Rightarrow \{-2c, -3c, c\}$

- The matrix is said to be in row-echelon form -
  1. 1st non-0 element in each row, c/d the leading entry is 1.
  2. Each leading entry is in a col. to the right of the leading entry in the prev. row.
  3. row with all 0 entries, should be present at the last.
  4. for a non-0 row, the leading entry in the row is the only non-0 entry in its column. $\Rightarrow$ Reduced row echelon

Eg. $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}$

Eg. $A_{ref} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ → but this is not in reduced ref.

⇓

$A_{rref} = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- How to find the sol^{ns} for such matrices.

$x_1 + 2x_2 = b_1$

$x_3 = b_2$ $\Rightarrow$ $x \begin{bmatrix} b_1 - 2c \\ c \\ b_2 \\ b_3 \end{bmatrix}$.

$x = b_3$. $x_2 = c$

- $Ax = b$ ⇒ A is rref.
  - → for some i^{th} row is 0 row, but $b_i \neq 0$. Then such a system does not have any solution.
  - → for every 0 row, we have $b_i = 0$, then
    - → if i^{th} col. has the leading entry of some row, we call that x_i, a dep var.
    - → if i^{th} col. does not have the leading entry of some row, we call such x_i an indep. variable.

$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}$

→ assign arbitary value to indep. var.

→ then find values of dep. var.

$x_1, x_2 \Rightarrow$ dep

$x_3 \Rightarrow$ indep

L5 Row Reducⁿ

- elementary row operations
  1. interchange 2 rows $R_1 \leftrightarrow R_2$
  2. scalar multiplicaⁿ of a row by a constant t $\overset{R_1/3}{\rightsquigarrow}$
  3. adding multiples of row to another row. $\overset{R_1-3R_2}{\rightsquigarrow}$
- use row reducⁿ to achieve the row echelon form
  1. use opⁿˢ to get 1 at $(0,0)^{th}$ locaⁿ.
  2. use opⁿˢ to get 0s below that 1 (above).
  3. continue the process, for other columns to get ref.
  4. to get rref, the entries above 1 also, should be 0.

Eg. $A = \begin{bmatrix} 2 & 4 & 1 \\ 3 & 8 & 7 \\ 5 & 6 & 9 \end{bmatrix} \xrightarrow[\text{ref}]{} \begin{bmatrix} 1 & 2 & 1/2 \\ 0 & 1 & 11/4 \\ 0 & 0 & 1 \end{bmatrix}$

det (A) = 70

$\downarrow$ rref

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

- $R_i \leftrightarrow R_j$ $\quad A \leftrightarrow B$ $\qquad \det(A) = -\det(B)$

$A \xrightarrow{R_i/c} B$ $\qquad \det(A) = c\det(B)$

$A \xrightarrow{R_i + cR_j} B$ $\qquad \det(A) = \det(B)$

- in this way, you can compute the determinant via the row reduc^n technique as well.

L6 Gaussian Elimina^n Method

- how to find solu^no
  - if A is invertible, solu^n is unique
    1. by cramer's rule
    2. adj. matrix to calculate A^{-1}.
  - better way was to use ref -
    1. find dep. & indep. var.
    2. assign values to indep. var, then, back calculate the values of dep. variables.
- The augmented matrix of the sys. $\underset{m\times n}{A}x = \underset{m\times 1}{b}$ is defined as the matrix of size $m\times n+1$. We denote the augmented matrix as $[A|b]$.

$$3x_1 + 2x_2 + x_3 + x_4 = 6$$
$$x_1 + x_2 = 2$$
$$7x_2 + x_3 + x_4 = 8$$

$$\Rightarrow \left[\begin{array}{cccc|c} 3 & 2 & 1 & 1 & 6 \\ 1 & 1 & 0 & 0 & 2 \\ 0 & 7 & 1 & 1 & 8 \end{array}\right]$$

- Gaussian elimina^n

consider a sys. of eq^{ns} $Ax = b$.

1. get the $[A|b]$
2. reduce A to get ref of $[A|b]$
3. Then, you get $[A|b] \longrightarrow [R|c]$

→ The solutions of $Ax = b$ are precisely the solutions of $Rx = c$.

4. form the eq^{ns} of $Rx = c$
5. find all the solutions of $Rx = c$, thus, obtain all the solutions of $Ax = b$.

Eg. $\left[\begin{array}{cccc|c} 3 & 2 & 1 & 1 & 6 \\ 1 & 1 & 0 & 0 & 2 \\ 0 & 7 & 1 & 1 & 8 \end{array}\right] \xrightarrow[\text{operations}]{\text{perform ref}} \left[\begin{array}{cccc|c} 1 & 2/3 & 1/3 & 1/3 & 2 \\ 0 & 1 & -1 & -1 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{array}\right]$

go to rref

$x_1 = 1$

$x_2 = 1$

$x_3 + x_4 = c$

$x_4 = c$

$x_3 = 1 - c$

$\Rightarrow \begin{bmatrix} 1 \\ 1 \\ 1-c \\ c \end{bmatrix}$

$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{array}\right]$

dep. var. — indep. var. (c)

- remember, if a i^{th} row has all 0 entries, but the corresponding b_i is non-0. That sys. of linear eq^n does not have any sol^n.

- homogenous sys. of linear eqns $\Rightarrow Ax = 0$
  1. one solⁿ is always present. 0 $\Rightarrow$ Trivial solⁿ
  2. There ∞ many solⁿ other than 0.
     $\Rightarrow$ Non-trivial solⁿ.
- In homogenous system of eqⁿs, if there are more variables than eqⁿs, then it is guaranteed to have non-trivial soluⁿ.